

Batch: A2 Roll No.: 16010322804

Experiment No. 1

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

TITLE: Analysis and representation of a discrete time system

AIM: 1) To represent different systems using the transfer function/ impulse response / Z domain.

2) To analyse discrete time systems in terms of the pole zero plot, magnitude and phase response impulse response, stability and type of the system.

OUTCOME: Analyse discrete time systems in time and Z-domain, Design and realize Digital Filters.

PROCEDURE:

1. Consider the various systems as given.
2. Represent the system in terms of transfer function / impulse response / Z domain.
3. Analyze the systems in terms of various parameters such as pole zero plot, magnitude and phase response impulse response, Minimum/maximum phase system, stability and type of the system.

SYSTEM I:

Pole located at $z = 1$, Zero located at $z = -1$

Analysis / Observations

Sr. No.	PARAMETERS	VALUES / OBSERVATIONS
1	Transfer function $H(z)$	$\frac{1+z^{-1}}{1-z^{-1}}$
2	Difference equation	$y(n) = x(n) + y(n - 1) + x(n - 1)$
3	Type of the filter based on impulse response FIR / IIR	IIR
4	Type of the filter based on the magnitude response (LPF, HPF, BPF, BSF)	High-Pass Filter (HPF)
5	Type of the filter based on the phase response (linear / non – linear)	Non-linear
6	Stability	Unstable
7	Effect on magnitude and phase response after adding additional pole at $z = 0$	The pole at the origin adds a one-sample delay (linear phase term $e^{-j\omega}$). Magnitude response remains unchanged, while the phase response decreases by an additional $-\omega$. Stability remains unchanged because the original pole at $z = 1$ is still on the unit circle.

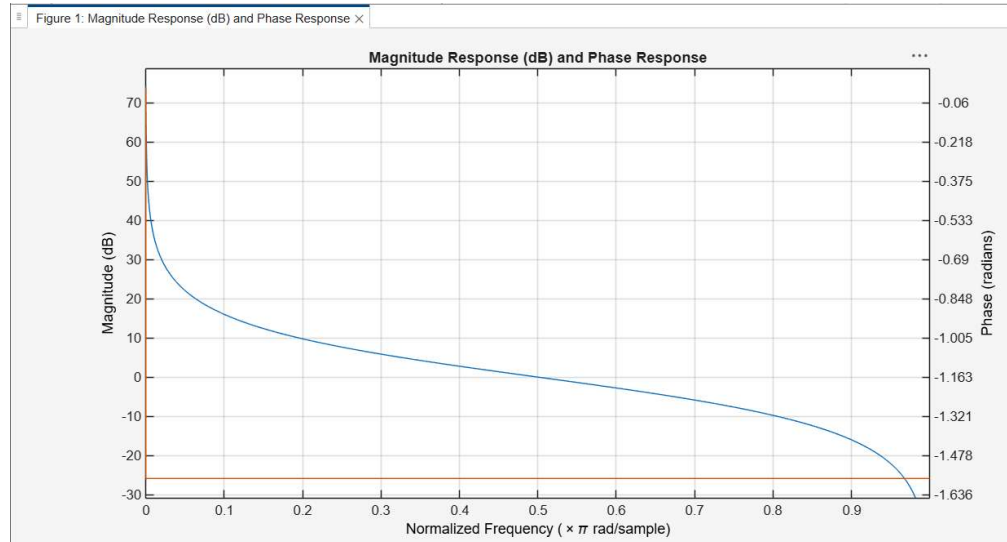
Department of Electronics and Telecommunication Engineering

Code:

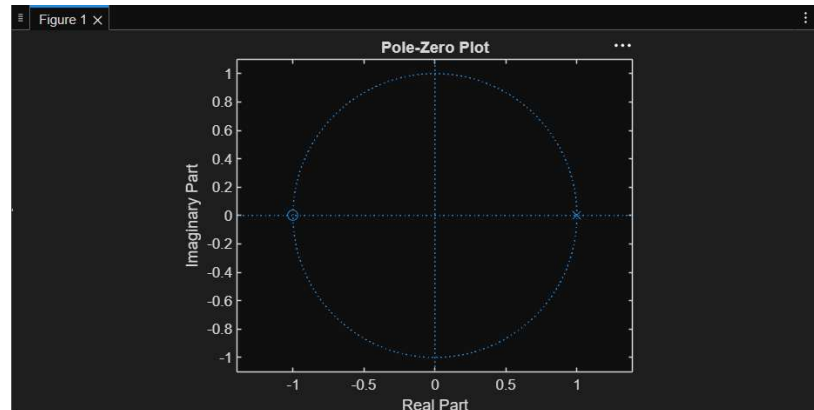
(Somaiya Vidyavihar University)

```
p1= [1]'; % Location of pole
z1= [-1]'; % Location of zero
[num, den]=zp2tf(z1,p1,1); % Pole-Zero to transfer function
zplane(z1,p1); % Plotting Pole-Zero
sys=tf(num,den)
ts=0.1
sys1=tf(num,den,ts, 'variable', 'z^-1')
fvtool(num, den) %Plotting Magnitude and phase response
```

Magnitude and phase spectrum:



Pole-Zero Plot:



(Somaiya Vidyavihar University)

SYSTEM II: Pole located at $z = 0$, Zero located at $z = 1$

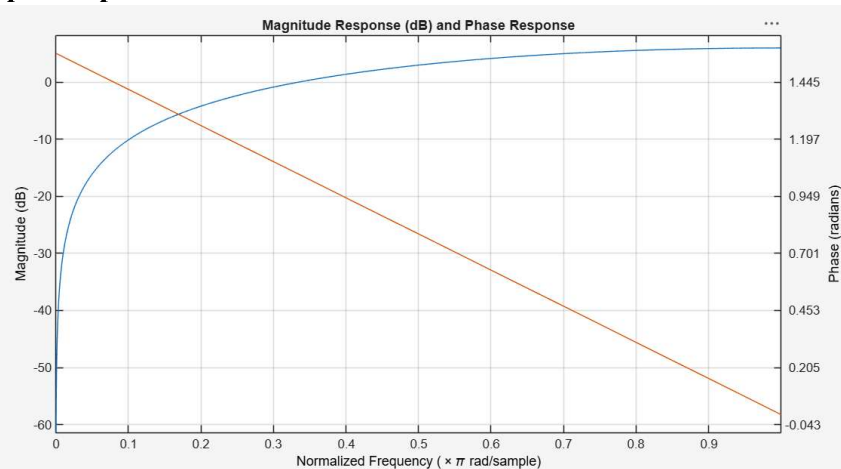
Analysis / Observations

Sr. No.	PARAMETERS	VALUES / OBSERVATIONS
1	Transfer function $H(z)$	$H(z) = 1 - z^{-1}$
2	Difference equation	$y(n) = x(n) - x(n - 1)$
3	Type of the filter based on impulse response FIR / IIR	FIR
4	Type of the filter based on the magnitude response (LPF, HPF, BPF, BSF)	High-Pass Filter (HPF)
5	Type of the filter based on the phase response (linear / non – linear)	Linear Phase
6	Stability	Stable

Code:

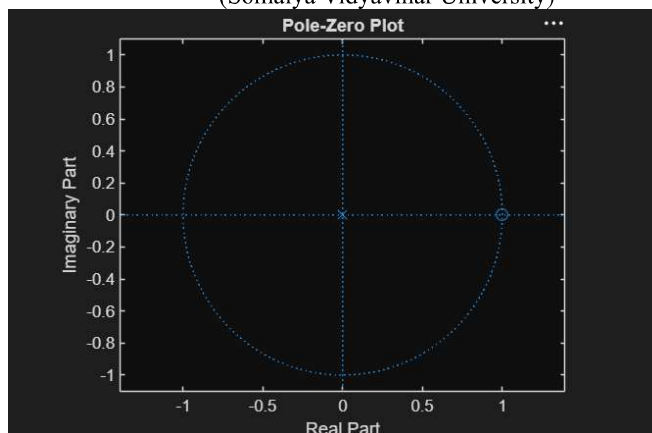
```
p1= [0]'; % Location of pole
z1= [1]'; % Location of zero
[num, den]=zp2tf(z1,p1,1); % Pole-Zero to transfer function
zplane(z1,p1); % Plotting Pole-Zero
sys=tf(num,den)
ts=0.1
sys1=tf(num,den,ts, 'variable', 'z^-1')
fvtool(num, den) %Plotting Magnitude and phase response
```

Magnitude and phase spectrum:



Pole-Zero Plot:

(Somaiya Vidyavihar University)



SYSTEM III: Pole located at $z = 1, -1$, Zero located at $1\angle 45^\circ, 1\angle -45^\circ$

Analysis / Observations

Sr. No.	PARAMETERS	VALUES / OBSERVATIONS
1	Transfer function $H(z)$	$H(z) = \frac{1 - \sqrt{2}z^{-1} + z^{-2}}{1 - z^{-2}}$
2	Type of the filter based on impulse response FIR / IIR	IIR
3	Type of the filter based on the magnitude response (LPF, HPF, BPF, BSF)	Band-Stop Filter
4	Type of the filter based on the phase response (linear / non – linear)	Non-linear phase
5	Stability	Unstable
6	Comment on the system after adding another zero located at $1\angle -45^\circ$	The zero at -45° becomes a double zero, producing a deeper and sharper notch at -45° . The phase changes, but the stability remains unchanged because the poles are unchanged.

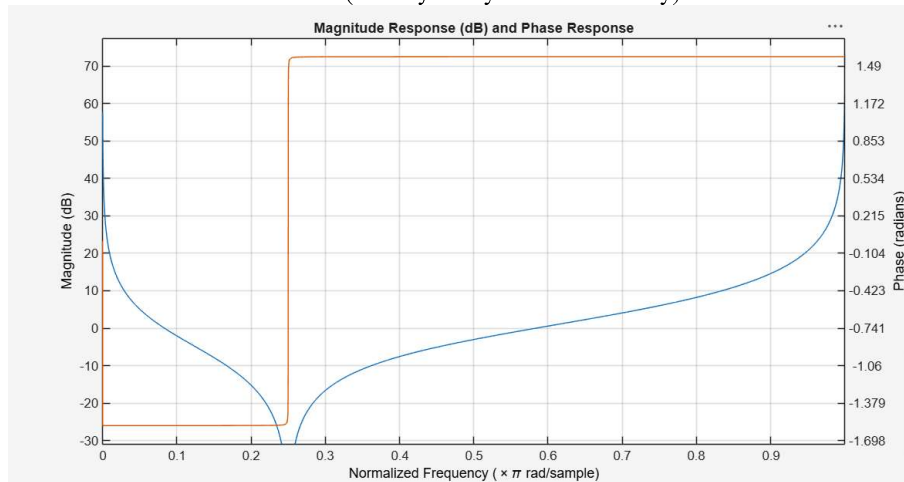
Code:

```

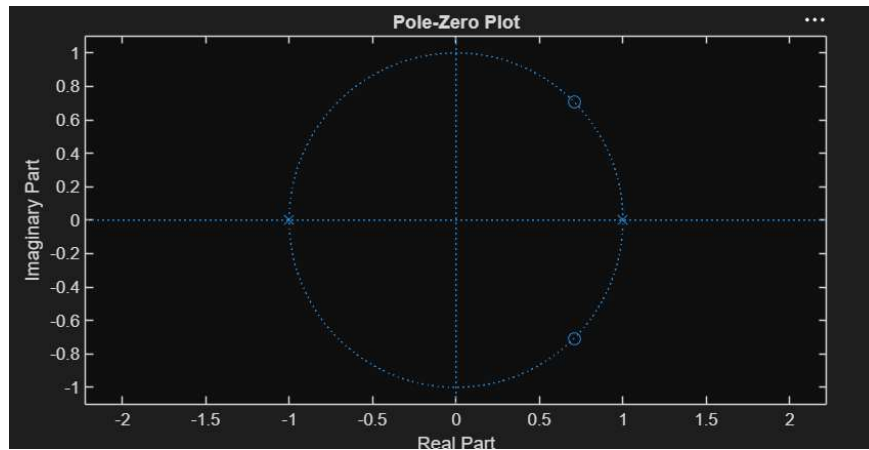
p1= [1, -1]'; % Location of pole
z1= [0.707+j*0.707, 0.707-j*0.707]'; % Location of zero
[num, den]=zp2tf(z1,p1,1); % Pole-Zero to transfer function
zplane(z1,p1); % Plotting Pole-Zero
sys=tf(num,den)
ts=0.1
sys1=tf(num,den,ts, 'variable', 'z^-1')
fvtool(num, den) %Plotting Magnitude and phase response
  
```

Magnitude and Phase Spectrum:

(Somaiya Vidyavihar University)



Pole-Zero Plot:



SYSTEM IV: $h(n) = (0.8)^n$ $0 \leq n \leq 7$

Analysis / Observations

Sr. No.	PARAMETERS	VALUES / OBSERVATIONS
1	Transfer function $H(z)$	$H(z)=1+0.8z^{-1}+0.64z^{-2}+0.51z^{-3}+0.40z^{-4}+0.32z^{-5}+0.26z^{-6}+0.20z^{-7}$
2	Type of the filter based on impulse response FIR / IIR	FIR
3	Type of the filter based on the magnitude response (LPF, HPF, BPF, BSF)	Low-Pass Filter
4	Type of the filter based on the phase response (linear / non – linear)	Non-linear phase
5	Stability	Stable

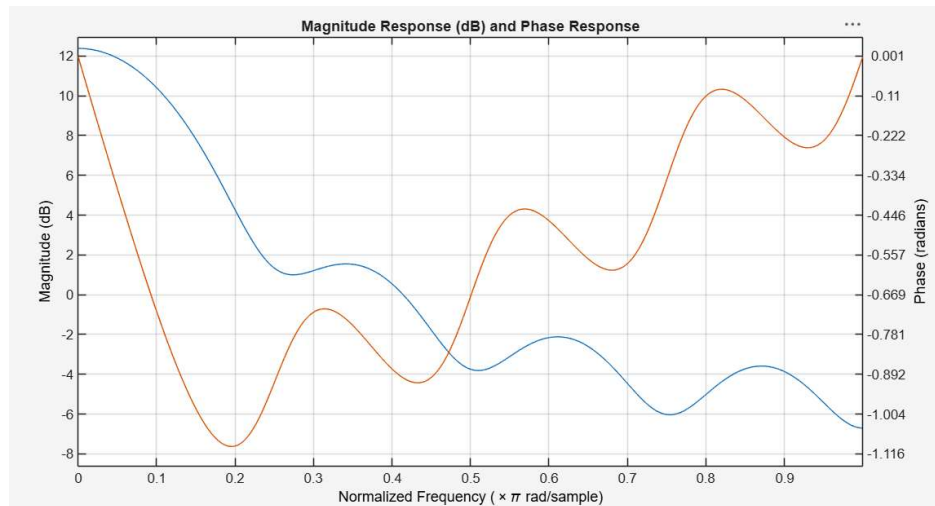
(Somaiya Vidyavihar University)

Code:

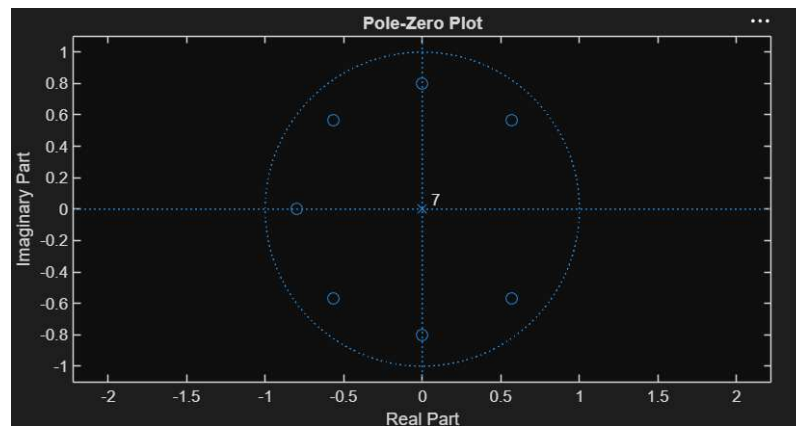
```
n = 0:7;
h = 0.8.^n;
num = h;
den = [1 zeros(1,length(h)-1)];
[z1,p1] = tf2zp(num,den);
zplane(z1,p1);
sys = tf(num,den)
ts = 0.1;
sys1 = tf(num,den,ts,'variable','z^-1');
fvtool(num,den)
```

% sample index n = 0 to 7
% impulse response $h(n) = (0.8)^n$
% numerator = impulse response coefficients
% denominator = 1 (FIR filter)
% get zeros and poles from num/den
% plot pole-zero diagram
% create transfer function
% sampling time
% discrete TF in z^{-1} form
% view frequency responses

Magnitude and phase spectrum:



Pole-Zero Plot:



SYSTEM V: $H(z) = 1 + z^{-4}$

Analysis / Observations

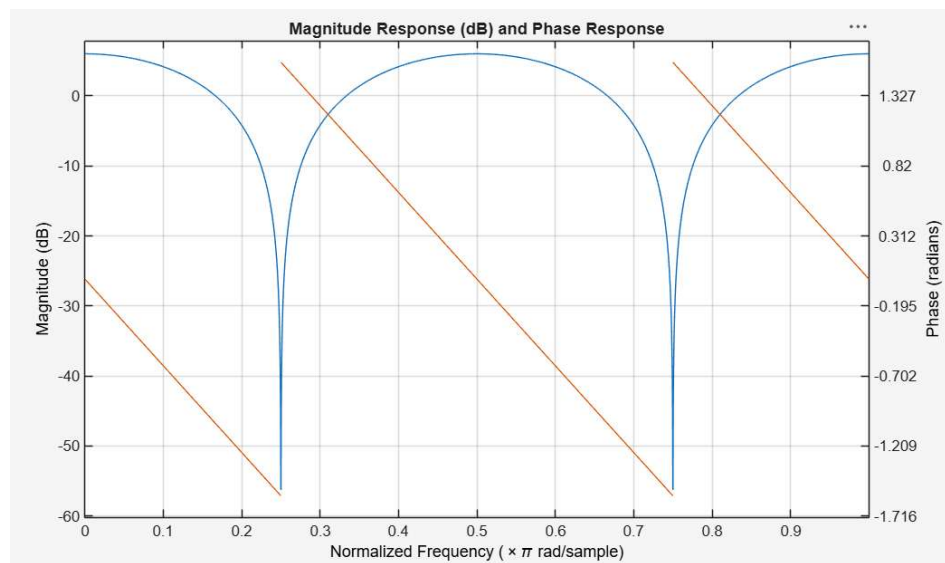
Sr. No.	PARAMETERS	VALUES / OBSERVATIONS
1	Impulse response $h(n)$	$h(n) = \delta(n) + \delta(n - 4)$
2	Type of the filter based on impulse response FIR / IIR	FIR

3	Type of the filter based on the magnitude response (LPF, HPF, BPF, BSF)	Low pass Filter
4	Type of the filter based on the phase response (linear / non – linear)	Linear phase
5	Stability	Stable
6	Suggest a suitable transfer function to obtain the same magnitude response as above but null located at $\omega = 0$	$H(z)=1-z^{-4}$

Code:

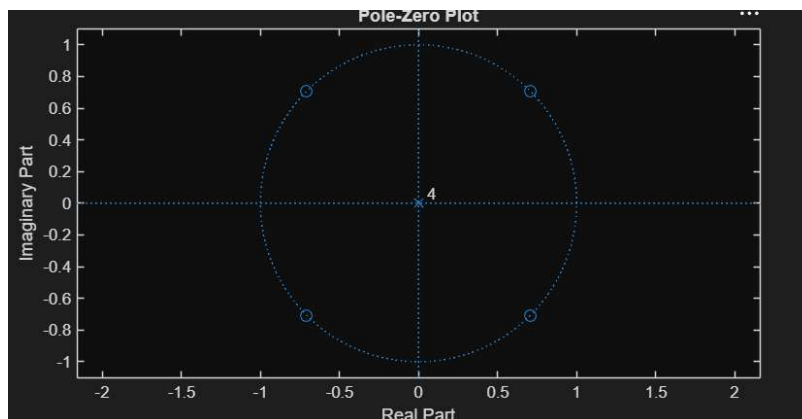
```
num= [1 0 0 0 1]; % co-efficient for 1+z^-4
den= [1 0 0 0 0]; % denominator = 1, padded to match length
[z1, p1]=tf2zp(num, den); % get numerator and denominator from poles and zeros
zplane(z1,p1); % Plotting Pole-Zero
sys=tf(num,den)
ts=0.1
sys1=tf(num,den,ts, 'variable', 'z^-1')
fvtool(num, den) %Plotting Magnitude and phase response
```

Magnitude and phase spectrum:



Pole-Zero Plot:

(Somaiya Vidyavihar University)



SYSTEM VI: $H(z) = \frac{0.5 + z^{-1}}{1 + 0.5z^{-1}}$

$$1 + 0.5z^{-1}$$

Analysis / Observations

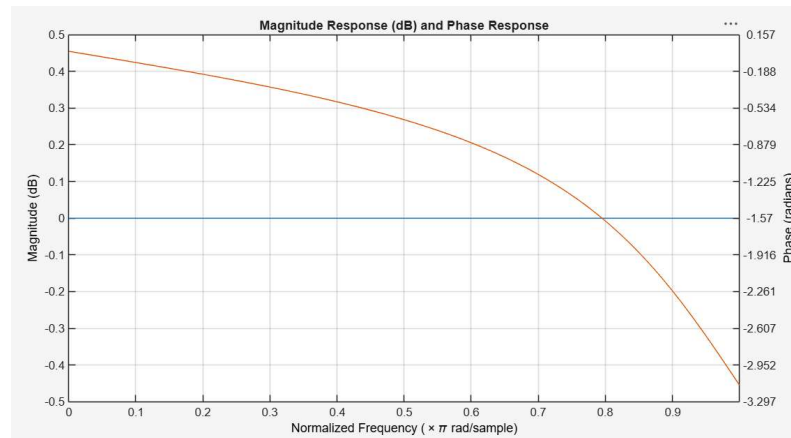
Sr. No.	PARAMETERS	VALUES / OBSERVATIONS
1	Difference equation	$y(n) + 0.5y(n-1) = 0.5x(n) + x(n-1)$
2	Type of the filter based on impulse response FIR / IIR	IIR
3	Type of the filter based on the magnitude response	All-Pass Filter
4	Type of the filter based on the phase response (linear / non – linear)	Non-linear phase
5	Stability	Stable
6	Comment on the transfer function	It is a first-order all-pass filter. It changes the phase of the signal while keeping the magnitude response constant at all frequencies. The zero at $z = -0.5$ is the reciprocal of the pole at $z = -2$, a characteristic property of all-pass filters.

Code:

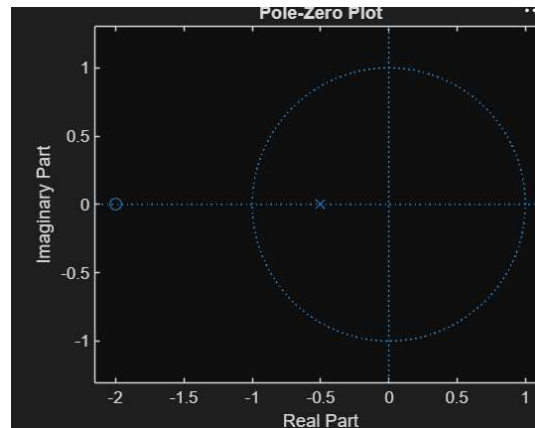
```

num= [0.5 1]; %co-efficient for 1+z^-1
den= [1 0.5]; %denominator = 1, padded to match length
[z1, p1]=tf2zp(num, den); % get numerator and denominator from poles and zeros
zplane(z1,p1); % Plotting Pole-Zero
sys=tf(num,den)
ts=0.1
sys1=tf(num,den,ts, 'variable', 'z^-1')
fvtool(num, den) %Plotting Magnitude and phase response
  
```

Magnitude and phase spectrum:



Pole-Zero Plot:



CONCLUSION:

In this experiment, I successfully carried out to represent various discrete-time systems using transfer functions, impulse responses, and the Z-domain. Different systems were analyzed using pole-zero plots, and their magnitude response, phase response, stability, and filter characteristics were studied. The relationship between pole-zero locations and the system behavior, such as FIR/IIR nature, LPF/HPF/BPF/BSF characteristics, linear/non-linear phase response, and stability, was verified through MATLAB simulations. Thus, the objectives of representing and analyzing discrete-time systems in both the time and Z-domains were successfully achieved.

Signature of faculty in-charge